

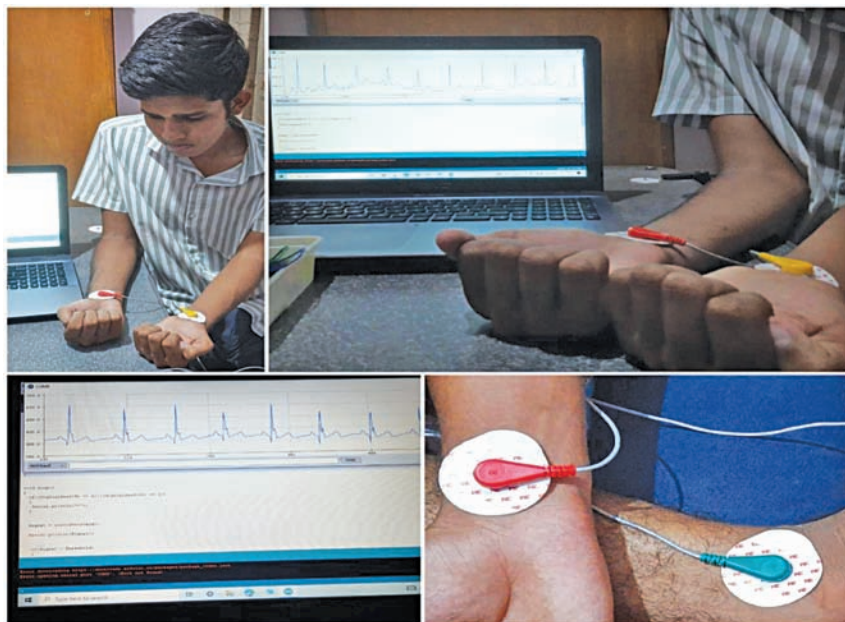


Fig. 8: Connections of the red, yellow, and green electrodes at RA, LA, and RL respectively



Fig. 9: ECG on a serial plotter

P, Q, R, S, T, and U intervals. Fig. 7 shows a complete waveform. The sinoatrial node (SA) is the pacemaker of the heart and produces the P wave. The QRS wave is produced by the AtrioVentricular (AV) node.

The P wave in an ECG complex indicates atrial depolarisation. The QRS is responsible for ventricular depolarisation, and the T wave shows ventricular repolarisation. QRS wave comprises a small Q wave, a high R wave, and a small S wave. The QRS segment offers information about the ventricular systole concerning the impulse propagation of the ventricles (Q wave). The R and S waves cause the transmission to the whole tissue.

Testing

As per the instructions shared, the electrodes are attached at the mentioned positions, as shown in Fig. 8. The Arduino is powered and a serial connection with the Arduino allows receiving the data at the baud rate of 9600—the rate of data transfer from Arduino to the laptop/workstation.

Pins 9 and 10 of the Arduino are used to receive the ECG sensor output, which can be printed using a serial plotter. Pin 11 is used for the buzzer operation. Whenever signal amplitude goes above the threshold level of 590 microvolts, the buzzer sounds, indicating hyperactivity. An example of the ECG graph is shown in Fig. 9. **EFY**

Dr Geetali Saha (left) is Faculty, Dept of Electronics & Communication, G.H. Patel College of Engineering & Technology (GCET), Anand, Gujarat, and Kushal Gadhi is a bio-med electronics enthusiast